

Recovery Is Part of Completion

Why backup existence is not enough for enterprise AI

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Executive summary

A system is not recoverable merely because a backup exists. Enterprise AI needs stronger proof: backup existence, integrity verification, restore testing, and verification of the restored state. This paper explains why recovery belongs inside the definition of completion for consequential agentic work.

The backup fallacy

A backup file can be corrupt, incomplete, stale, inaccessible, or impossible to restore in the environment where it is needed. Treating “backup exists” as “recovery works” converts an intention into an unsupported operational claim.

Four recovery predicates

A stronger recovery contract has four distinct predicates:

- **Backup exists.** The required recovery object is present.
- **Integrity verified.** Its identity and expected contents are independently checked.
- **Restore tested.** The restore procedure is actually executed.
- **Restored state verified.** The restored result matches the expected known-good state.

Skipping any step leaves a different class of uncertainty.

Recovery should be subject-bound

Recovery evidence should identify exactly what it can restore. A generic backup directory is not enough when multiple versions or environments exist. The recovery record should bind subject identity, backup identity, restore method, expected state, dependencies, and any limitations.

Test recovery before failure

The worst time to discover that a restore path is incomplete is during an outage or failed deployment. Periodic restore testing converts recovery from documentation into observed capability. For consequential changes, a narrower restore test can be part of the release gate.

Recovery and safe change

Safe change benefits from a verified predecessor and a proven restore path. If a successor fails, the system can return to the predecessor with less ambiguity. Recovery also supports experimentation by limiting the cost of failed changes.

Partial failure and ambiguity

Agentic workflows can fail after some side effects occur. In that situation, blind retry can duplicate actions. A resilient system classifies the state—completed, partial, not completed, or ambiguous—then reads back actual

side effects before deciding whether to retry, compensate, or roll back.

Recovery evidence in independent review

A reviewer should be able to inspect recovery proof without trusting the producer’s label. Useful evidence includes exact backup identity, integrity recomputation, restore logs or receipts, and the restored-state identity.

A practical enterprise checklist

Question	Expected evidence
What is backed up?	Exact subject/version binding
Is the backup intact?	Recomputed integrity evidence
Can it be restored?	Observed restore test
Did restore produce the right state?	Restored-state verification
What happens after partial failure?	Readback/compensation rules
Who can trigger recovery?	Explicit authority boundary

Limitations

No recovery plan can guarantee restoration under every catastrophe. Dependencies, credentials, third-party outages, and environmental changes can still block recovery. The discipline makes those dependencies visible and tests what is realistically testable.

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